

4.11 Key ideas

The complex and diverse phenomena of the natural world can be described in terms of a small number of key ideas in chemistry.

These key ideas are of universal application, and we have embedded them throughout the subject content. They underpin many aspects of the science assessment and will therefore be assessed across all papers.

These ideas include:

- matter is composed of tiny particles called atoms and there are about 100 different naturally occurring types of atoms called elements
- elements show periodic relationships in their chemical and physical properties
- these periodic properties can be explained in terms of the atomic structure of the elements
- atoms bond by either transferring electrons from one atom to another or by sharing electrons
- the shapes of molecules (groups of atoms bonded together) and the way giant structures are arranged is of great importance in terms of the way they behave

- there are barriers to reaction so reactions occur at different rates
- chemical reactions take place in only three different ways:
 - proton transfer
 - electron transfer
 - electron sharing
- energy is conserved in chemical reactions so can therefore be neither created or destroyed.